

Состязательные атаки

x - объект $f(x)$ - нейросеть

y_{true} - настоящий класс

$\ell(\cdot, \cdot)$ - ф-я потерь

$\tilde{x}$ - adversarial объект:

$$\begin{cases} \|\tilde{x} - x\|_{\infty} \leq \frac{\epsilon}{255} \\ f(\tilde{x}) \neq y_{true} \end{cases}$$

White-box атаки - архитектура известна

Black-box атаки - " " не известна

$f(\tilde{x}) = y_{target}$ - targeted атака

$f(\tilde{x}) \neq y_{true}$ - untargeted атака

FGSM (Fast Gradient Sign Method)

Targeted: $\tilde{x} = x - \epsilon \text{sign}(\nabla_x \ell(f(x), y_{target}))$

Untargeted: $\tilde{x} = x + \epsilon \text{sign}(\nabla_x \ell(f(x), y_{true}))$

I-FGSM (iterative FGSM)

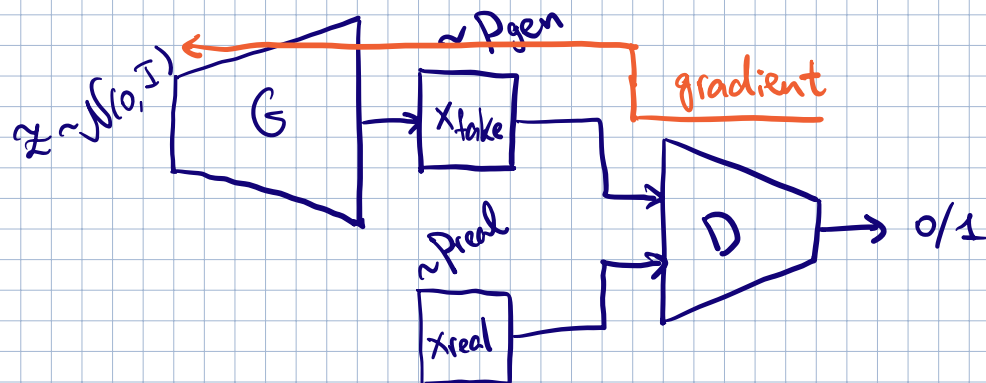
$x_0 := x$

for $t = 1 \dots T$:

$x_t := x_{t-1} - \frac{\epsilon}{T} \text{sign}(\nabla_x \ell(f(x_t), y_{target}))$

$\tilde{x} := x_T$

GAN (Generative Adversarial Nets)



$$\mathcal{L}_{GAN} = \mathbb{E}_{x \sim P_{real}} [\log D(x)] + \mathbb{E}_{z \sim \mathcal{N}(0, I)} [\log (1 - D(G(z)))] \rightarrow \max_D \min_G$$

cGAN (conditional GAN)

$$G(z) \rightarrow G(z, c)$$

$$D(x) \rightarrow \begin{cases} D_{GAN}(x) \\ D_{cls}(x) \end{cases}$$

Instance Norm
 $m(c), s(c)$

$\{c_1, \dots, c_k\}$
↓ embeddings

$\{e_1, \dots, e_k\}$

$$m(c_i) = \text{Linear}_1(e_i)$$

$$s(c_i) = \text{Linear}_2(e_i)$$

Pix2pix (Img2Img)

(x_s, x_T) - training dataset

1) MSE $G: \text{Img} \rightarrow \text{Img} \quad \|G(x_s) - x_T\|_2^2 \rightarrow \min_G$

2) MSE + GAN $\mathcal{L}_{GAN}(G, D) + \text{MSE}(G) \rightarrow \max_D \min_G$

$$L_{GAN} = \mathbb{E}_{x \sim P_{real}} [\log D(x)] + \mathbb{E}_{x \sim P_{gen}} [\log (1 - D(x))] \rightarrow \max_D$$

$$L_{GAN} = \int \log(D(x)) \cdot P_{real}(x) dx + \int \log(1 - D(x)) P_{gen}(x) dx$$

$$= \int [P_{real}(x) \log D(x) + P_{gen}(x) \log(1 - D(x))] dx$$

$$P_{real}(x) \cdot \log c + P_{gen}(x) \log(1 - c) \rightarrow \max_c \quad \left| \frac{d}{dc} \right.$$

$$\frac{P_{real}(x)}{c} - \frac{P_{gen}(x)}{1 - c} = 0$$

$$c P_{gen}(x) = P_{real}(x) (1 - c)$$

$$D^*(x) = c = \frac{P_{real}(x)}{P_{real}(x) + P_{gen}(x)}$$

- $x: P_{gen}(x) \approx 0 \Rightarrow D^*(x) \approx 1$
- $x: P_{gen}(x) \gg P_{real}(x) \Rightarrow D^*(x) \approx 0$

$$L_{GAN} = \int \log(D^*(x)) \cdot P_{real}(x) dx + \int \log(1 - D^*(x)) P_{gen}(x) dx =$$

$$= \int \log\left(\frac{2 P_{real}(x)}{P_{gen}(x) + P_{real}(x)}\right) P_{real}(x) dx + \int \log\left(\frac{2 P_{gen}(x)}{P_{gen}(x) + P_{real}(x)}\right) P_{gen}(x) dx$$

$$= \underbrace{- \int \log 2 (P_{real}(x) + P_{gen}(x)) dx}_{2 \log 2} =$$

$$= KL(p_{real} \parallel \frac{p_{gen} + p_{real}}{2}) + KL(p_{gen} \parallel \frac{p_{gen} + p_{real}}{2}) - 2 \log 2 = 2 \underbrace{JS(p_{real}, p_{gen})} - 2 \log 2$$

